

VANTARA

Revenue Leak Analysis

BUSINESS REQUIREMENT DOCUMENT (BRD)

Supply Chain, Delivery & Customer Retention

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1. Executive Summary

Order volume grew every year, but revenue fell about 56% from the September 2017 peak to December 2017. Where is Vantara losing money and customers, and what should be done about it?

Vantara is a supply chain and logistics company selling across USCA, Europe, LATAM, Africa and Pacific Asia. Leadership saw rising order volume that did not translate into matching profit, inconsistent delivery, and no clear view of true customer loyalty. This project takes 180,519 raw order-line records through cleaning, relational modeling, SQL analysis, an independent Excel cross-check and an executive Power BI dashboard to answer the question above with verified numbers.

180,519	63,629	20,652	\$32.76M	2015–17
order-line records	orders	customers	revenue	full years

2. Business Problem

Leadership observed the following pattern over 2015–2017:

- **Volume up, revenue not.** Orders kept increasing, but revenue growth slowed and in some months reversed sharply.
- **Unreliable delivery.** Delivery times were inconsistent across shipping modes, and a high share of orders arrived late.
- **Unknown loyalty.** It was unclear how loyal the customer base really was, or which segments deserved more investment.
- **Unresolved revenue.** A large share of orders sat in unresolved statuses (pending payment, on hold, fraud review), making true revenue hard to pin down.

As Business Analyst, the task was to investigate the order, customer, product and delivery data, find what was really happening, and turn it into practical recommendations leadership can act on.

3. Objective

Study Vantara's order, customer, product and shipping data to answer one core question: "**Where is Vantara losing money and customers, and what should be done about it?**" The solution is a complete, documented analysis pipeline: data cleaning, relational database, SQL analysis, Excel cross-check and an interactive Power

BI dashboard.

4. Why This Dataset

The dataset was chosen because it contains a genuine diagnostic problem rather than a pre-labeled one:

- **Hidden problem.** Order volume rises every year, so a shallow read says the business is healthy. Only by separating revenue per order, order status mix and delivery performance does the Q4 2017 revenue collapse appear underneath.
- **A wrong answer already in the data.** An early reference figure put customer loyalty at 76%. It was recalculated from first principles in SQL, DAX and Excel, giving 56.98%, before being trusted.
- **Full analyst toolchain.** BRD first, then Python cleaning, PostgreSQL star schema, SQL analysis, Excel cross-check and Power BI, with a Decisions Log recording each judgment call.
- **Rewards depth.** Each insight has a first answer and a truer one underneath. The project is built to stop at the second, not the first.

5. Stakeholders

Stakeholder	What they care about	Decision they make
CEO	Overall growth and direction of the business	Where to invest or pull back
CFO	Revenue, profit margin, confirmed vs at-risk revenue	Forecasts, pricing and promotion limits
VP of Operations	Delivery speed, late shipments, shipping mode reliability	Carrier and shipping-mode fixes
VP of Customer Success	Retention, repeat purchase rate, segment behavior	Win-back and loyalty programs
Marketing Team	Which regions, products and segments to prioritize	Campaign and inventory focus

6. Key Business Questions

#	Business question	KPI used	Where answered
Q1	Why did revenue drop sharply in late 2017 while order volume stayed high?	Total Revenue, Revenue per Order	SQL trend queries, Overview page
Q2	Which categories and regions bring the most and least profit?	Profit Margin %	SQL, Excel pivots
Q3	How reliable is delivery, and which shipping modes are late most often?	Late Delivery Rate, Avg. Fulfilment Time	Delivery Performance page
Q4	How much revenue is confirmed (Complete/Closed) vs pending, on hold or fraud risk?	Order Status Mix	SQL, Excel summary
Q5	How loyal are customers, and does loyalty differ by segment?	Repeat Customer Rate	Customer Insights page

#	Business question	KPI used	Where answered
Q6	What data-backed actions protect profit and improve retention?	All of the above	Findings report

7. Project Scope

In scope

- Order, customer, product, shipping and payment data for full calendar years 2015–2017
- Year-over-year and month-over-month performance comparison
- Profitability by region, category and customer segment
- Delivery performance by shipping mode and order status
- Customer retention and segment-level behavior
- PostgreSQL database, SQL queries, Excel model, Power BI dashboard and a findings report

Out of scope

- **January 2018 data.** One partial trailing month cannot be compared fairly with full years and would distort month-over-month and year-over-year trends. It is kept in raw tables for traceability only and excluded from every KPI, trend and dashboard view.
- Real-time or live data tracking (historical analysis only)
- Direct changes to Vantara's operations, pricing or systems (this project recommends; operations decides)
- Customer-level personal targeting (analysis stays at segment and aggregate level)

8. Data Source

Source: *DataCo Smart Supply Chain for Big Data Analysis* (Kaggle), 180,519 order-line records. The raw data arrived as one flat table and was restructured into a star schema.

Table	Contents	Size
fact_orders	Order date, ship date, sales, profit, discount, quantity, status, shipping mode, delivery status, late flag, fulfilment time, payment type	63,629 orders
dim_customers	Customer ID, segment (Consumer / Corporate / Home Office), city, state, country	20,652 customers
dim_products	Category, department, product name, price	118 products
dim_location	City, state, country, market, region	Market / region level
dim_date	Calendar attributes for trend analysis	2015–2017

Source link: kaggle.com/datasets/shashwatwork/dataco-smart-supply-chain-for-big-data-analysis

9. Methodology

Step	Tool	What was done
1. Clean	Python (Pandas)	Fixed data types, removed irrelevant and sensitive columns, standardized text, corrected malformed records, excluded partial Jan 2018 at load time

Step	Tool	What was done
2. Model	Python, PostgreSQL	Split the flat file into 1 fact + 4 dimension tables with primary and foreign keys; validated row counts and keys
3. Analyze	SQL	Joins, CTEs and window functions for revenue trends, profit by category, late delivery, repeat customers, payment mix, order status
4. Cross-check	Excel	Independent pivot tables mirroring SQL, plus a what-if model for the revenue impact of fewer late deliveries
5. Visualize	Power BI (DAX)	Three-page dashboard (Overview, Delivery Performance, Customer Insights) on the Vantara brand theme
6. Recommend	Report	Plain-English findings, business impact and prioritized actions

10. Key Performance Indicators

KPI	Definition	Why it matters
Total Revenue	Sum of sales, 2015–2017 (headline: \$32.76M)	Overall financial performance
Profit Margin %	Total profit divided by total revenue (headline: 12%)	How much revenue becomes profit
Late Delivery Rate	Share of deliveries flagged late; stated with its grain (order line or order)	Reliability of shipping
Avg. Fulfilment Time	Mean days from order to delivery (headline: 3.47 days)	Speed to customer
Repeat Customer Rate	Customers with more than one order divided by all customers (headline: 56.98%)	Loyalty and retention
Order Status Mix	Revenue share by status (Complete, Closed, Pending, On Hold, Fraud)	Confirmed vs at-risk revenue

11. Assumptions & Decisions Log

Decision	Reason
Exclude Jan 2018 from all analysis	Partial month distorts MoM and YoY comparison
Define Repeat Customer Rate as customers with 2+ orders; replace the 76% reference figure with 56.98%	Recalculated independently in SQL, DAX and Excel; all three agree
State grain next to every delivery metric	SQL counts order lines (54.83% late); Power BI counts orders. The two agree directionally but not in raw count
Describe the revenue decline as September peak (\$1.03M) to December (\$453K), about 56%	Avoids overstating it as a full-quarter average
Remove sensitive and irrelevant columns at cleaning	Analysis stays at segment and aggregate level

12. Risks & Open Items

Item	Impact	Action
Repeat-customer figure differs on dashboard (56.98% card vs 63.51% narrative)	Conflicting message to leadership	Lock one definition as the single source of truth
Late delivery counted at different grains (SQL vs Power BI)	Headline numbers do not match	Document grain; automate SQL-vs-DAX reconciliation
Revenue map renders U.S. states only	Hides non-U.S. revenue (England, Isle of France)	Add world map or market toggle
Q4 2017 root cause not yet isolated	Cause could be pricing, promotions or data capture	Cohort and pricing analysis for Oct–Dec 2017
Shipping-mode service levels need verification	Premium tiers may not perform as assumed	Confirm late rate by shipping mode in SQL before recommending any tier

13. Deliverables

- Cleaned dataset: fact_orders, dim_customers, dim_products, dim_location, dim_date
- PostgreSQL schema with referential integrity
- SQL analysis query file
- Excel workbook: pivot tables, scenario model, executive summary
- Power BI dashboard: Overview, Delivery Performance, Customer Insights (published live)
- Assumptions & Decisions Log
- Findings & Recommendations Report

14. Success Criteria

- A clear, data-backed explanation of where revenue and profit came under pressure
- At least five concrete, prioritized recommendations for leadership
- Every headline KPI verified in at least two tools
- A dashboard leadership can explore independently
- Documentation clear enough for a new analyst to follow without extra explanation

15. Document Control

Version	Change
2.1	Original BRD: Smart Supply Chain Intelligence Platform
3.0	Renamed to Revenue Leak Analysis; added executive summary, question-to-KPI map, decisions log, risks and open items

End of Document — Vantara Business Requirement Document v3.0 (Final)